



Department of Computer Science and Engineering

Academic Year 2021 - 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering

Course Code & Title: CS8451 & Design and Analysis of Algorithms

Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / Closest-Pair and Convex Hull Problem
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Minute Paper
- **Justification:**

Closest-Pair and Convex Hull problem is one of the important topics, repeatedly asked in university questions. Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concept of closest pair and convex hull algorithms taught in class. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

- **Time Allotted for the Activity:** 5 Minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the concept of closest pair and convex hull algorithms discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. The students also write about the specific topic which is needed to be discussed or clarified in the further classes. This activity shows whether the students can able to understand the specific topic and their involvement the particular class. The activity of minute paper conducted was shown in Figure 1.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PSO1	PSO2	PSO3
CO1	3	2	1	1	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4
	3	2	1	1
Justification for correlation	Convex hull and closest pair problem is a basic concept necessary to solve problem.	These concepts are used in analysis of engineering problems	To analysis complex problems and represent using these concepts	To provide the solution for complex problem using convex hull concept.
	PO10	PSO1	PSO2	PSO3
	2	2	1	1
	Able to represent the solutions for the closest pair and convex hull problems	Able to use these concepts in the applications of Information Technology	Able to use these algorithmic concepts for developing reliable IT Solutions	Convex hull concept is helpful in solving real world problems in Research.

• **Images / Screenshot of the practice:**

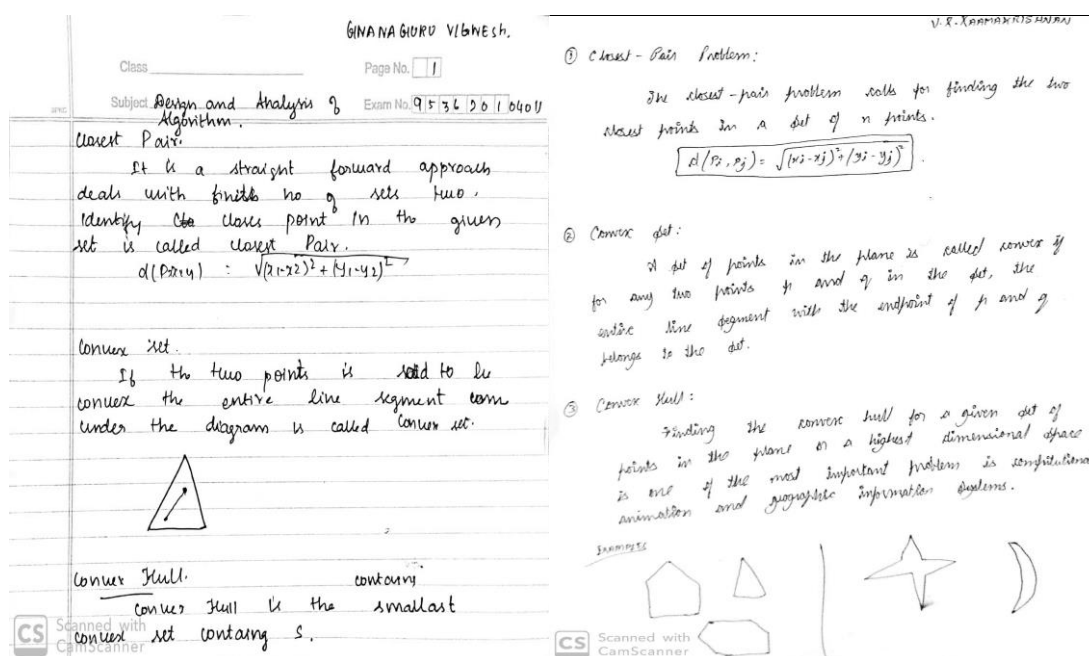


Figure 1: Minute Paper Activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Use of minute papers in the classroom actually increases attentiveness in the class and also improves the students' skill of writing and critical analysis of the topics.

- This activity reflects the understanding and involvement of the students in the particular topic.

❖ ***Benefit of the practice:***

- Students were keenly participated in this activity.
- From this activity, each student can refresh the closest pair and convex hull concept and understood the topic clearly while the information is discussed and recalled.

❖ ***Challenges faced in implementation:***

- Some students are hesitated and lack to share the concepts that they learnt in the class.
- Slow learners were felt difficult to write the key points in the topic.

References:

- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
- ❖ <https://www.unl.edu/gradstudies/current/teaching/minute>
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

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